

TechSched Business Proposal

Item	Value
Product	TechSched — intelligent dispatch for on-site repair
In one line	Lets a 5—20 technician repair company run routine dispatch with zero human involvement, and hands the decisions that do need a person to that person together with the reasoning
Team / team code	(to be filled in)
Code repository	https://github.com/yuyuchen0204/techsched-hackathon
Live system	https://byyy.com/techsched/
Technical design document	docs/TechSched-Technical-Design.pdf (architecture, agent workflows, algorithms, security, deployment, evaluation)
Submission date	(to be filled in)

This document complements the technical design document. It answers why the problem is worth solving, for whom value is created, how that value is measured, and whether it can be deployed. Technical detail — architecture diagrams, algorithms, tool definitions, prompts and tests — lives in the technical design document and is cited here only where it supports a business claim.

About the numbers. Two kinds appear. Those marked **[measured]** come from the evaluations and tests in the code repository and can be reproduced. Those marked **[assumed]** are our estimates of the target customer's operations, not yet validated in the field; they are stated explicitly so the reasoning can be checked and corrected.

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1. One Business Story

The problem we are solving. A repair company with 10 technicians takes 20–30 jobs a day. What actually consumes people is not producing the schedule but what happens after it: a technician calls in sick, a customer chases, the previous job overruns. Every one of those changes means somebody has to work out again who can go, whether sending them will make someone else late, and how many phone calls that takes.

Who experiences it. A company this size normally has no dedicated dispatch role. The work falls to the owner or an administrator, who takes calls while looking for a technician on a spreadsheet, judges from experience whether they will make it in time, and leaves no record of what was changed or why.

What the agents create. TechSched takes over three things: understanding the customer, deciding whether a job can actually be scheduled, and working out who a change would affect. Routine orders run end to end with nobody involved. When something changes, the system re-plans within the authority the company has configured and presents the result as one card carrying a decision score, the list of affected orders and a line-by-line diff — so the dispatcher's job becomes approve or reject, not re-derive. When the situation is beyond that authority or has no feasible answer, the system does not fail silently: it hands over a case with the actions it already tried.

The outcomes we expect. Less time spent on dispatch, faster response to changes, a higher on-window arrival rate, and a share of human involvement that can be tracked month by month. On the part the system can already measure, **the number of rule violations in committed plans is always zero** [measured].

2. Problem and Opportunity

2.1 Target Customer and Stakeholders

Stakeholder	Role in this process	What they care about
Owner / administrator (dispatch)	Takes bookings, assigns work, handles changes	Whether any job cannot be placed today, which ones need my decision, whether a change will upset a customer
Field technician	Carries out the visit	Where the next job is, when to leave, the unit number, why a job was inserted
Reporting customer	Requests the repair, waits	When somebody will come, whether they can fix it, whether it can be moved or expedited
Owner (as a business)	Looks at results	Labour spent on dispatch, on-time rate, cost per job, whether volume can grow without new headcount

Target profile: a Singapore-based on-site repair company with 5–20 technicians, 15–40 jobs a day, covering two or more trades. That size falls within Singapore's official definition of a small or medium enterprise (annual turnover not exceeding S\$100 million, or no more than 200 employees).

2.2 Today's Process and Its Pain Points

Pain point	How it shows up	Business consequence
Feasibility is estimated	At scheduling time there is no way to confirm the technician will arrive in time or hold the right skill	Late arrivals and repeat visits; customer complaints
Change has no boundary	Several customers' appointments get moved to rescue one urgent job, purely on judgement	Damage to long-standing customers, with no way to review it afterwards
Decisions are not traceable	A change is made and that is the end of it; no version, no reason	Disputes cannot be settled; experience cannot be accumulated
Incomplete information	Missing unit number, unclear problem description	A technician makes a wasted trip
Quality cannot be reproduced	Dispatch quality equals one person's condition that day	The company cannot grow past that person

2.3 Why It Is Worth Solving

- **It happens constantly.** Change is not an edge case. Our evaluation scenarios contain six classes of event in a single day — new insertions, paid expedites, technician cancellations and more [measured: 6.1 events per random seed]. In real operations, leave, reschedules and traffic delays are routine.
- **The cost is immediate.** One badly handled change costs a customer their time, a technician half a day, and possibly a bad review.
- **Without solving it the company cannot grow.** Dispatch capability is tied to one individual; at 20 technicians that person becomes the constraint.

2.4 The Opportunity

Field-service management software is a growing segment; several research firms put the global market for 2026 between US\$5.6 billion and US\$6.1 billion, with a compound annual growth rate of roughly 14–15%. But the pricing and implementation complexity of those systems target mid-size and large enterprises, and **repair companies with 5–20 technicians are still working with a phone and a spreadsheet**. Large language models have, for the first time, made "turn what the customer said into a structured work order" cheap enough for a product aimed at that size of customer to be viable.

3. The Solution in Business Terms

3.1 What the System Does

Three front ends over one back office:

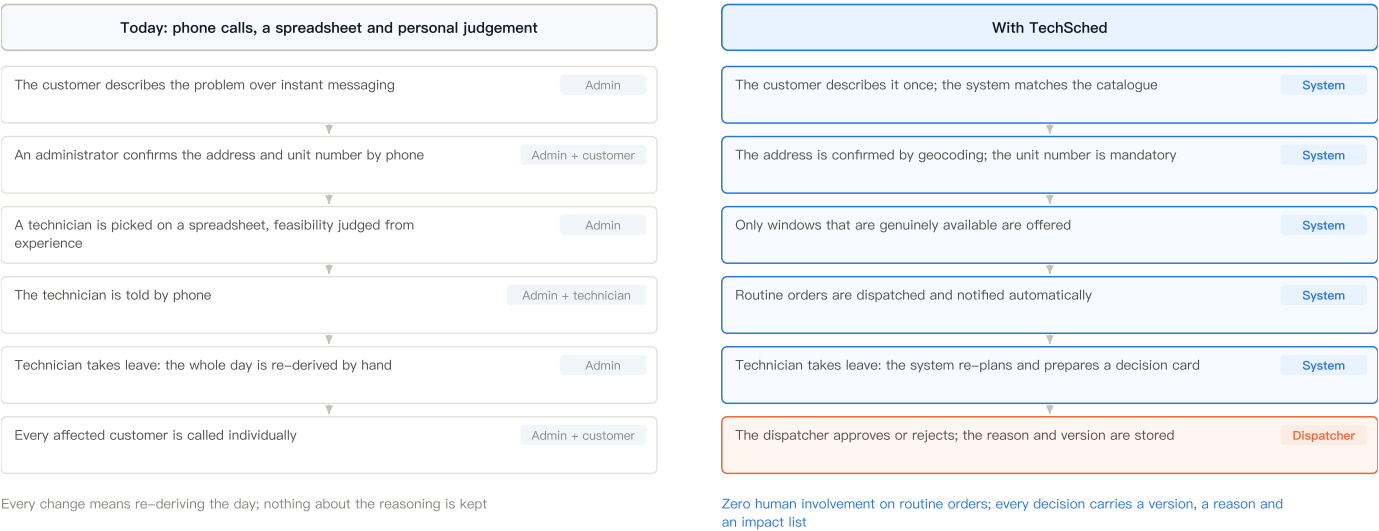
Front end	Used by	What it does
Customer app	Reporting customers	Book by conversation, confirm the address, choose a window that is genuinely available , track progress, expedite or reschedule
Technician app	Field technicians	See the next job (with unit number and customer history), report execution state, take leave, declare a rest
Dispatch console	Owner / administrator	See the day's schedule and risks, approve the plans that need a decision, work the human queue

Behind them are three agents and a deterministic scheduling engine. **The key product judgement is that the model understands and chooses, while rules decide feasibility and authority.** Skill matching, repair duration, whether something may execute automatically and how many orders may be moved are all determined by code and by the company's own configuration, not by the model.

3.2 Before and After

The dispatch process: today vs. with TechSched

The left column is how the target customer works today (an assumption, not a measurement); the right column is the same process once this system is running



The left-hand steps are an assumption about current practice and were not timed on site; the right-hand steps describe this system's actual behaviour

Figure 1 The same process under both ways of working. In the right-hand column only the last step keeps a person — and that step is approve or reject, not re-derive.

3.3 Where the Human Stays

The boundary of automation is explicit and configurable, not a matter of the model's discretion.

Where every dispatch decision ends up

The routing rules live in config/policy.yaml and are not decided by the model; the boundary of automation is configurable and auditable by the business

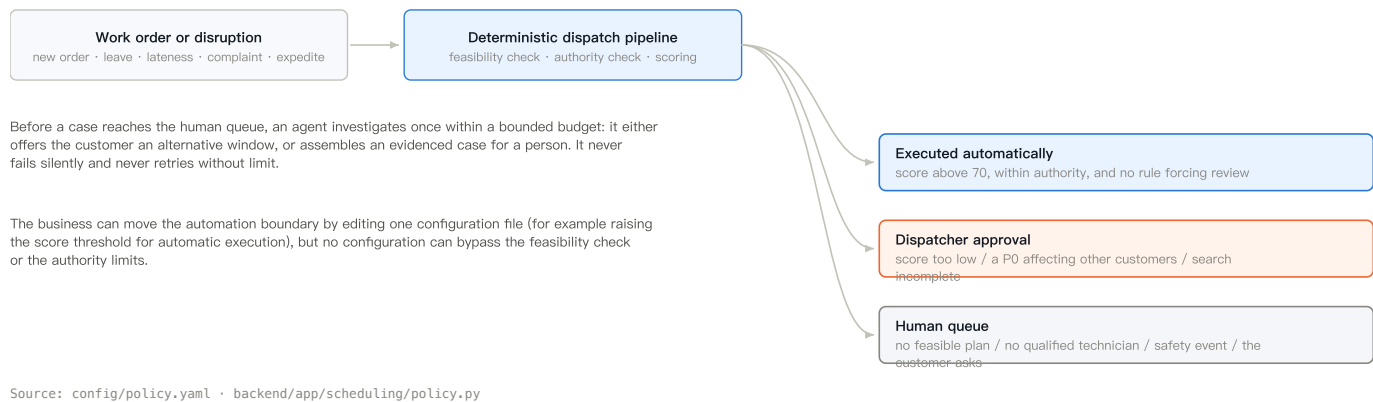


Figure 2 The three destinations of a dispatch decision and the conditions that send it to each. The rules live in one configuration file; the company can move the boundary, but no configuration can bypass the feasibility check or the authority limits.

For the owner this means three things:

1. **The degree of automation is adjustable.** To be more conservative, raise the score threshold for automatic execution and more plans go to review.
2. **High-risk decisions always carry a signature.** Whenever an urgent job needs to move another customer's appointment, dispatcher approval is mandatory no matter how high the score [measured: guaranteed by the order of the policy ruling and covered by tests].
3. **The system never pretends it coped.** With no feasible plan it produces an evidenced case for a person rather than going quiet or retrying indefinitely.

4. Business Value

Value is set out along the six dimensions. Items marked [measured] are behaviours the system already has and that can be verified; items marked [assumed] are the operational effects inferred from them, to be validated in a pilot.

Dimension	What the agents do	Effect on the business
Productivity	A routine order runs from the customer's description to a dispatched job with nobody involved [measured: the fast path completes within one request and creates no human task]	Dispatch staff move from handling every job to handling only exceptions [assumed: frees roughly 60% of dispatch time]
Cost	Dispatch decisions are made by deterministic code; the model is used only for conversation and classification [measured]	Model cost grows with the number of conversations , not with schedule complexity — more and harder jobs do not raise unit cost
Revenue	Expediting is a chargeable feature with transparent consequences: before payment the customer is told how many normal jobs will move and whether dispatcher confirmation is needed [measured]	A revenue line tied directly to the scheduling capability [assumed]
Service	The customer gets "09:30–11:00, Bala expected at 10:33" instead of "this afternoon", and an explanation whenever they are moved [measured]	Fewer chasing calls and complaints [assumed]
Risk	Hard constraints and authority are re-checked by an independent validator; over-authority plans have no approval control at all [measured: zero violations in committed plans across both evaluations]	Fewer promises the company cannot keep, and fewer disputes arising from them
Scale	The same rules handle 12–, 20– and 32–order scenarios, with a mean solve time of 0.9 milliseconds [measured]	Growth in volume does not require a proportional increase in dispatch staff

5. Impact and Measurable Outcomes

5.1 KPI Design

Each KPI states where in the system it is read from, so a pilot can take the figures directly rather than relying on impressions.

Business outcome	KPI	Baseline (today)	Target	Where it is read
Reduce manual work	Share of routine orders needing a person	[assumed] 100%; every job passes through a person	< 10%	Agent scorecard <code>autonomy.rate_pct</code>
Improve productivity	Time to handle one change	[assumed] about 15 minutes (re-derive plus phone calls)	< 3 minutes (approve one card)	Approval event timestamps (plan generated → approved)
Improve customer service	On-window start rate for the original window	[assumed] no record today; judged from complaints	Visible monthly and improving	<code>on_window_rate</code> from evaluation and execution events
Reduce risk	Rule violations in committed plans	[assumed] unknowable; no validation exists	Always 0	<code>hard_or_authority_violations</code> [measured: already 0]
Improve response	Urgent jobs left unserved	[assumed] no definition today	Declining monthly	<code>urgent_unserved</code>
Support growth	Technicians supported per dispatcher	[assumed] about 10	20 or more	Technician count ÷ dispatch FTE

5.2 A Worked Value Model

For a company with 10 technicians. **Every input states its source**, so the company can substitute its own figures and recompute.

Input	Value	Source
Technicians	10	Assumed (target profile)
Jobs per day	25	Assumed (same density as the 8-technician, 20-order demonstration scenario)
Working days per month	26	Assumed
Dispatch effort today	2.5 hours a day (about 0.3 FTE)	Assumed
Share of that spent on routine dispatch	60%	Assumed
Human involvement in routine orders	0	Measured (the fast path creates no human task)
Time to handle one change	15 minutes → 3 minutes	The first assumed; the second an estimate of approving one card

Result: dispatch effort falls from 2.5 hours a day to about 0.9, roughly 42 hours a month. At an assumed fully loaded dispatcher rate of S\$18 an hour that is about **S\$750 of labour released per month** — and, more importantly, that time was previously unavailable for taking bookings or looking after customers.

This is a **model, not a promise**. Its purpose is to break the claim into inputs that can each be challenged. The first task of a pilot is to replace this table with real data and, after 4–8 weeks, read the KPIs in 5.1 against it.

5.3 Evidence Already Measured

The figures below come from the offline evaluations in the code repository and can be reproduced with `POST /api/evaluations`. They run on synthetic worlds and are used for like-for-like comparison under a common baseline; they are not a quantified claim about real business returns.

V2 controlled experiment: two dispatch strategies over the same initial schedule and event sequence
10 random seeds × 6.1 events per seed = 61 events; route provider fixture; policy version 2026-09-14-v2

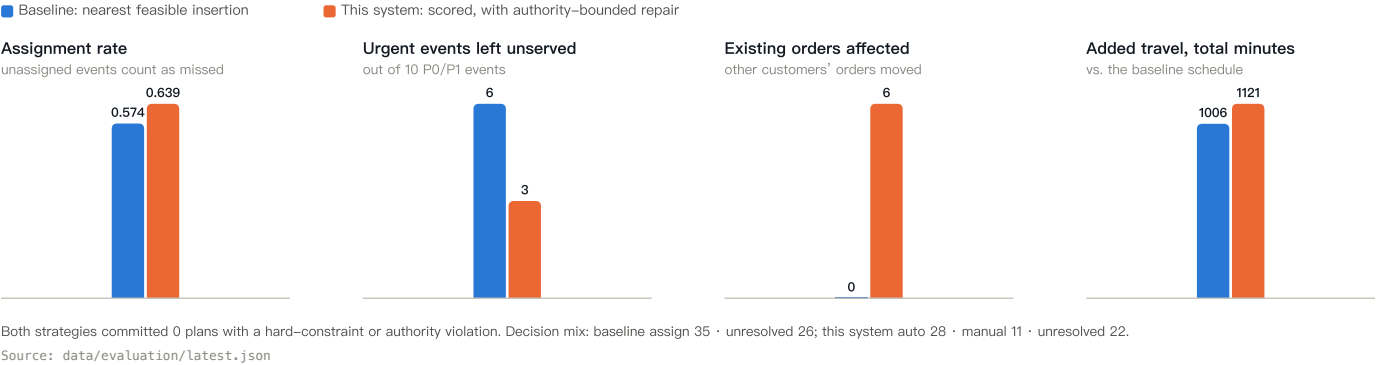


Figure 3 Nearest feasible insertion compared with this system's strategy over the same initial schedule and event sequence. Each of the four panels covers one measure.

Claim verified	Evidence
Better service of urgent jobs	Urgent events left unserved fall from 6/10 to 3/10
The cost is quantified and shown	Total travel +115 minutes and 6 existing orders affected (3 technician changes, 205 minutes of shift) — the system shows this trade-off to the dispatcher rather than hiding it
The rules are not broken	Across both strategies and three load scenarios, hard-constraint and authority violations in committed plans are zero
Performance is not the constraint	Mean solve time 0.9 ms, maximum 33 ms, against a 3–5 second budget
The agents do not run away	Under a real model, five tasks each used 4–5 tool calls and all ended in an evidenced human case; no loops, no exhausted budgets
The system actually runs	128 backend tests and 42 three-end assertions all pass; the deployment is reachable

6. Feasibility

6.1 Data and Knowledge Needed to Maintain It

This is the part of adoption cost most often underestimated, so it is set out separately.

What is needed	Who maintains it	Effort	How we handle it
Repair catalogue (trade, problem, complexity, standard duration)	The company's technical lead	Compile once, then top up quarterly	One CSV; 46 records already cover 10 trades. The import reports the actual headers, the field mapping and any bad rows, and conflicting duplicate keys abort the import rather than being guessed
Technician roster and skill levels	The owner	Enter once, update on staff changes	One JSON or form. Skill level is a hard dispatch condition, so an error shows up as a job that cannot be assigned, never as a silent downgrade
Business rules (priority, authority, approval threshold, rest limits)	The owner	Set once, rarely changed	Centralised in one configuration file; changing a rule needs no code change
Customers and history	Accumulated automatically	None	Completed jobs become history and preferences

When the catalogue is missing the system refuses to create orders rather than falling back to an invented one [measured]. That is a deliberate choice: better not to work at all than to promise a time derived from a guessed duration.

6.2 Systems, Tools and APIs

Capability	Current implementation	Exercised live
Language model	Replaceable provider layer: OpenAI-compatible endpoint / Anthropic SDK / offline mock	Running in production (model identifier <code>global.anthropic.claude-sonnet-4-5-20250929-v1:0</code> , via an OpenAI-compatible gateway)
Road network and travel times	OSRM (public server or self-hosted)	Yes
Address resolution	OneMap (Singapore postal codes and HDB blocks) → Nominatim	Yes, with a real OneMap account
Notifications	In-app, labelled as simulated	Not yet on a real SMS channel (a pilot item)
Payment	Simulated payment events	Not yet on a real gateway (a pilot item)

All three providers return the same data structure, so **changing model or routing supplier requires no change to business logic** — which is both a cost-control mechanism and a hedge against supplier lock-in.

6.3 Human-in-the-Loop and Escalation

Human involvement is part of the product, not a fallback, with explicit triggers and a hand-over standard:

- **Three sources:** the customer asks, policy requires it, or an agent escalates. All three enter one queue that can be taken, replied to and closed; while a customer has an open case the assistant pauses and makes no scheduling commitment [measured].
- **Hand-over standard:** an agent escalation must carry evidence references, the actions already attempted and a suggested next step; missing any one of them is not a valid escalation [measured: enforced by the tool contract].
- **No repeated interruptions:** an open case of the same category is merged into, with evidence appended and urgency raised, rather than duplicated.

For the business this means **the length of the human queue is a manageable operating metric**, not a pile of uncategorised exceptions.

6.4 Deployment and Operating Needs

Item	Status
Deployment shape	Single-process backend plus static frontend, nginx reverse proxy and systemd; live at https://byyy.com/techsched/
Runtime status	https://byyy.com/techsched/health returns the active model, routing, geocoding and policy version, and contains no secrets
Deployment action	One script: install dependencies → build the frontend → sync static files → restart the backend → health check
Runs offline	The default configuration is an offline mock model with a built-in travel matrix, so it runs straight after cloning — a customer can try it without connecting any external service
Model cost structure	Scheduling itself makes no model calls; the model is used only for conversation and complaint classification. Cost grows with conversations, not with job volume or schedule complexity

Actual token consumption is not yet instrumented. That is the first thing to add in week one of a pilot, and no estimate is offered here.

6.5 Risk and Exposure

Risk	Assessment	Mitigation
Uncontrolled model output	The model cannot set a priority, a duration or an authority limit — no tool exists for it [measured]	Structural isolation rather than prompt instructions; customer text is declared to be data
External service outage	Model or routing unavailable	Automatic degradation, labelled in the interface: the model falls back to the rule implementation, routing to a straight-line estimate [measured]
Promising the impossible	Giving a customer a time that cannot be met	Only genuinely feasible windows are offered, and the choice is re-validated on submission rather than silently accepted [measured]
Safety events (gas, electric shock)	The customer describes a dangerous situation	Scheduling stops, verified official numbers are shown, a critical case is created; the system does not report on the customer's behalf and does not dial [measured]
Data compliance	Customer addresses and phone numbers	Currently synthetic data; before production a retention policy and role-based permissions are required (see 7.3)
Supplier lock-in	Dependence on one model vendor	The provider layer is replaceable; three implementations already exist

7. Scalability

7.1 More Users and Transactions

- **There is headroom in the algorithm.** Across three load scenarios (12 / 20 / 32 orders) the mean solve time is 0.9 milliseconds and the maximum 33, against a 3–5 second budget [measured]. The constraint today is not solving.
- **There is a known limit in the architecture.** Deployment is single-process behind a global lock, a deliberate trade-off at prototype stage. Moving to multiple processes means replacing that lock with database-level concurrency control — a known engineering item, not an architectural rewrite (see 9.L6 of the technical design document).
- **The cost side is favourable.** Dispatch consumes no model calls, so growth in job volume does not raise model cost.

7.2 Cross-functional Expansion

The same structure — feasibility check, authority boundary, human approval — transfers to scheduling problems beyond repair: cleaning and installation, equipment inspection, home services. What has to be replaced is the catalogue and the skill definitions, not the scheduling engine.

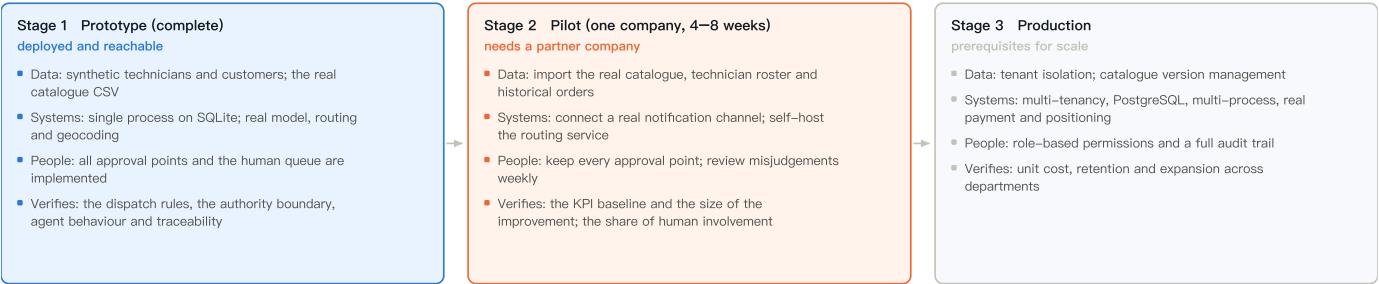
A nearer step is expanding within one customer: parts inventory (check at dispatch whether the part is on the van) and time-and-billing (execution facts are already recorded).

7.3 Monitoring, Security and Governance

Dimension	Today	Needed before production
Observability	Three layers of record: orchestration stages, step-by-step agent traces, the schedule version chain; every decision replays to a version and a reason [measured]	Central logging and alerting
Operating quality	Agent scorecard: autonomy rate, cost per task, share of futile calls, hand-over completeness [measured]	Alert thresholds
Authentication	A demonstration identity selector, not an authentication system	Real authentication and role-based permissions
Audit	The decision chain is fully retained	Retention policy, export and compliance audit
Multi-tenancy	Not implemented	Data isolation and per-company billing

7.4 From Prototype to Production

The path from prototype to production
What has to be added at each stage, and what that stage lets you verify



The system is at stage 1 today and is already running online (see chapter 7 of the technical design document). Stage 2 requires no rewrite: the catalogue, the roster and the business rules are all configuration, and connecting a real channel means swapping a provider implementation.

The timing and scope of stages 2 and 3 are planning assumptions and have not been carried out

Figure 4 What has to be added at each of the three stages, and what each stage lets you verify. The system is at stage 1 today and already running online.

Stage 2 requires no rewrite: the catalogue, the roster and the business rules are all configuration, and connecting real notification and payment channels means swapping a provider implementation. That is why every external dependency was collected into the provider layer.

8. Commercial Model and Route to Market

8.1 Pricing

Model	Description	Fits
Per-technician subscription	A fixed monthly fee per technician, covering dispatch, the customer app and the technician app	The primary model: customer size tracks value, and technician count is a number the company already knows
Share of expedite revenue	A share of what customers pay to expedite	Tied directly to the value we create; expediting only works because the system can prove a slot is genuinely earlier
One-off onboarding	Catalogue compilation, roster import, rule configuration	Lowers the barrier to first deployment

Pricing should be set after a pilot; no figure is given here, because without a real KPI baseline any price is a guess.

8.2 Route to Market

1. **One pilot customer, 4–8 weeks.** The objective is not a signed contract but a real baseline and improvement figure for the KPI table in 5.1.
2. **Lead with the catalogue.** Compiling the catalogue has value to the company in its own right (standard job times, a basis for quoting) and is a precondition for the system, which makes it a good first deliverable.
3. **Keep every approval point.** The pilot should not chase a higher automation rate; first let the dispatcher build trust in the system's judgement, reviewing misjudgements weekly.
4. **Review on the same metrics.** Autonomy rate, share of human involvement and on-time rate, read weekly, become the basis for deciding whether to move to production.

9. Claims and Demonstration Evidence

The demonstration video and this document must tell the same story. The table maps each business claim to the specific moment in the demonstration and to where it can be verified in the system.

Claim in this document	Evidence in the demonstration	Verifiable in the system
Routine orders need no human	The customer books in the app → confirms → the order goes straight to assigned; on the timeline no other technician's task has changed	Order status <code>ASSIGNED</code> , with no agent task created
High-risk decisions always carry a signature	A technician takes leave → the urgent job needs to move one normal order → the review card appears, and cannot execute automatically even with a score above the threshold	The policy ruling is <code>manual</code> ; the review queue
The cost of a change is transparent	The review card shows the decision score, 1 of 5 affected, and a line-by-line diff (one order re-assigned from A to B)	The candidate plan's diff table and affected list
Payment only buys what the rules allow	The customer expedites → before payment they are told how many normal jobs move and whether confirmation is needed; when a paid slot is not actually earlier the option is not shown	The expedite trial and the policy ruling
The system never fails silently	In the scarce scenario an order that cannot be placed → the agent investigates → offers alternative windows or escalates with evidence	The agent reasoning timeline and the human queue
Every decision can be replayed	The reasoning timeline shows, step by step, why this step, what was called, what came back and how long it took	The three layers of record and the schedule version chain

ACTIVE RISKS3

P2 approaching deadline wo_039

- waiting for your approval

P0 technician cancelled wo_039

- waiting for your approval

P3 unassigned eta unknown wo_039

- waiting for your approval

Review queue (1)Human queue (1)Agents (0)

Target wo_039 P0

Lock damaged needs replacement · Punggol Waterway Point · window 10:30–10:40

base v8 · expires 10:46

1 Faster Response · Less Disruption · Balancedpolicy: manualPENDING_REVIEW

TARGET START10:36TECHNICIANDevi NairTRAVEL / WAIT20 / 0 min

DECISION SCORE74.35 (min of changed · >70 auto)1 / 5 (movable: P2, P3)TRAVEL Δ30 min

Devi Nair starts wo_039 at 10:36 (travel 20 min, wait 0 min). Affects 1 order(s): wo_012 tech_04→tech_08 at 11:00. Lowest changed-assignment score 74.3 (threshold >70). Policy: manual — P0 affecting 1 undeparted P2/P3 orders: dispatcher approval required; decision_score 74.35 > 70.

order	pri	change	technician	start	Δ min	counts
012	P3	technician	tech_04 → tech_08	11:00		✓
015	P3	travel only	tech_08	13:30		
019	P3	travel only	tech_04	15:00		

show full diff & scores

Approve & commitRejectRecompute

Figure 5 The review card from the demonstration: the target order and its priority, decision score 74.35 against a threshold of 70, 1 of 5 affected, a line-by-line diff, and approve / reject / recompute. This is what "high-risk decisions always carry a signature" looks like in practice.

10. Assumptions and Limitations Register

10.1 Assumptions Used in This Document

ID	Assumption	Where it is used	How to validate
A1	The target customer handles 25 jobs a day with 10 technicians	5.2 value model	The pilot customer's actual volume
A2	Dispatch takes about 0.3 FTE today, 60% of it routine	5.2 value model	A week of time records before the pilot
A3	One change currently takes about 15 minutes	5.2 value model	Timing on site before the pilot
A4	A fully loaded dispatcher rate of S\$18 an hour	5.2 value model	The customer's actual cost
A5	The target customer currently dispatches by phone and spreadsheet	Chapter 2	Pilot customer interviews
A6	Expediting is acceptable to customers as a chargeable feature	Chapters 4 and 8.1	Actual usage and payment rate during the pilot

10.2 Current Limitations of the System

ID	Limitation	Consequence
L1	Demonstration data is synthetic; notifications and payment are simulated	Real channels must be connected before a pilot
L2	The catalogue currently holds 46 records across 10 trades	Coverage directly determines the share handled automatically
L3	Routing excludes live traffic; the adjustment factor is an engineering default	Needs calibrating against actual departure and arrival times
L4	The solver is heuristic with no guarantee of optimality; single day, single region	Multi-day and multi-region scheduling are not implemented
L5	Single-process deployment; the identity selector is not authentication	Multi-process operation and real authentication are needed before production
L6	Evaluation data comes from synthetic worlds	Not a quantified claim about real business returns

Submitted together with `docs/TechSched-Technical-Design.pdf`. Every conclusion marked *[measured]* can be reproduced from the repository with `scripts/run_tests.sh` and `POST /api/evaluations`; every figure can be regenerated from repository data with `scripts/handbook/make_figures.py`.